# Sahaj Bhadja

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## Summary

Software engineer with 2+ years building maintainable applications, REST APIs, and data-backed systems. Strong in Python, Java, JavaScript/TypeScript, OOP, testing, debugging, documentation, and secure coding.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*M.S. in Computer Science* *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Tech, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*B.Tech. in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Eng., OS, OOP, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*
- Designed SQL-backed platform applications for 500+ users, tuning indexed queries, validation logic, and background jobs across FastAPI services to cut dashboard latency from 1.2s to 350ms.
- Built fault-tolerant orchestration and integration workflows for 1,000+ AI tasks and recurring updates, strengthening debugging speed, release reliability, and maintainable behavior across production operations.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Engineered Flask services, REST APIs, and React workflows handling 10,000+ daily requests, integrating secure uploads, grading history, and instructor-facing features for a high-volume web application.
- Designed concurrent batch-processing for 1,000+ uploaded files using multithreading, AWS RDS, and CI/CD-backed releases, reducing feedback turnaround by 60% while maintaining stable APIs under peak demand.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*
- Engineered API and ETL integrations with Python, SQL, AWS Glue, and S3, improving response time by 30% while supporting scalable source-to-target workflows and maintainable backend components.
- Maintained 20+ automated tests and Jenkins pipelines with PyTest and Mockito, increasing CI reliability by 68% and strengthening validation, deployment, debugging, and development discipline.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*
- Built full-stack product features with REST APIs, WebSockets, and responsive interfaces, supporting 2M+ synchronized records across modular workflows and reliable end-to-end application behavior.
- Automated Docker-based build and test flows with Jenkins CI/CD, cutting release time by 50% and improving deployment consistency, troubleshooting speed, and engineering collaboration across product environments.

**LST Capitals** **India**
*Web Developer Intern* *November 2022 – January 2023*
- Developed backend APIs for a booking platform covering reservations, scheduling, status updates, and payment workflows across 50,000+ records through maintainable Flask services.
- Mapped 12+ integration points and implemented payment and synchronization flows with Flask, Firebase, and webhooks, reducing manual payment handling by 70% and improving transaction reliability.

## Skills

**Languages**: Python, Java, JavaScript, TypeScript, SQL, C++, Bash
**Application Development**: object-oriented programming, design patterns, REST APIs, Flask, FastAPI, Node.js, Express, JSON
**Testing & Quality**: unit testing, integration testing, debugging, validation logic, code reviews, documentation, secure coding
**Platforms & Data**: Windows application exposure, PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, AWS RDS, S3
**Delivery**: Git, Jenkins, CI/CD, Docker, Agile/Scrum, stakeholder communication, troubleshooting

## Projects

**Chat2SQL** Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and structured data management through a unified Java and TypeScript app.

**CPU-scheduling Simulator** Live | *C++, React, EJ-2*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ with C++ logic, metrics, and Gantt visualization across 1,000+ simulated process combinations.

**HMIS** Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Built a healthcare operations platform with dashboards, scheduling, and alerts, reducing administrative workload by 30% through TypeScript services, structured workflows, monitoring, and responsive views.